

Overview

A verification-centered path from charge to coupled fields

THE BOUNDARY APPLIES TO EVERY PAGE

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person. The learner works only with the explicitly pictured, current-limited battery investigations. Deenergize before changing a circuit. The adult facilitator inspects the setup and controls stop decisions; adult status does not imply electrical qualification.

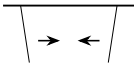
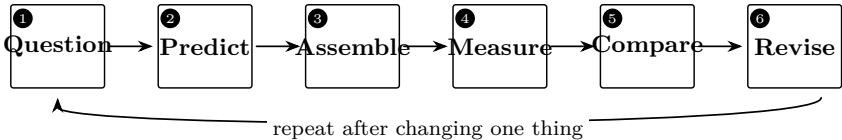
What this course is trying to prove

The course does not ask a learner to accept a named effect because a diagram looks persuasive. Each lesson asks a question, changes one variable, captures an observation or measurement, and decides whether the evidence supports the prediction. A successful apparatus is evidence only when the record says what was held constant and what would count against the claim.

Predict first

Opening question. Which is stronger evidence: “the coil worked,” a meter reading of 0.42 V, or a table showing that the reading changed when distance changed? What extra information would make your choice repeatable by another learner?

The loop used in every investigation



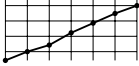
Ask before touching

What outcome would distinguish attraction from repulsion? Name it before preparing the tape.



Capture a quantity

Record distance, deflection, voltage, current, time, turns, or another quantity with its unit.



Look for a pattern

A trend, repeated null result, or disagreement is more informative than one dramatic trial.

CHECKPOINT 0 — READY TO BEGIN

The learner can state the voltage limit, identify the person who may stop the trial, show how power is disconnected, and write a prediction that could be wrong. If any part is missing, remain on the Safety sheet.

Overview

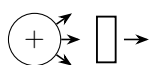
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ChatGPT edition — the ten-lesson
evidence staircase

Course sequence: each claim earns the next question

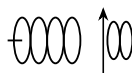
Lesson	Question	Evidence to preserve	Diagnostic checkpoint	Record artifact
1	Charge	Attraction or repulsion across repeated tape preparations; can motion versus approach distance.	Can the learner separate charge transfer from polarization?	Sign-comparison matrix.
2	Fields	Direction and magnitude of test-object deflection at five or more locations.	Do arrows describe a positive test charge, rather than visible threads in space?	Annotated field map.
3	Ohm's law	Matched voltage and current pairs with one resistor held constant.	Does V/I remain approximately stable within measurement resolution?	Data table and graph.
4	Electromagnet	Pickup or deflection versus current or turn count, changing only one.	Can the learner distinguish current, turns, and core material?	Controlled-comparison table.
5	Induction	Meter direction and magnitude while motion direction or speed changes.	Is a changing magnetic situation required?	Motion/response sequence.
6	Transformers	Primary and secondary turns with measured voltage ratio.	Does the ratio predict direction and approximate size of change?	Ratio calculation with uncertainty.
7	Capacitors	Charge/discharge voltage at fixed time intervals using only the bounded low-voltage setup.	Can stored energy be distinguished from a source still connected?	Time series and safe-discharge check.
8	Resonance	Response across a controlled frequency sweep in the supplied low-energy arrangement.	Is the peak repeatable and broader than reading noise?	Sweep plot with peak band.
9	Coupling	Received signal versus coil distance or orientation.	Does one geometric change predictably alter transfer?	Geometry sketch plus readings.
10	Spark gap	Explanatory comparison of historical discharge apparatus; no learner construction or energization.	Can the learner explain why the course boundary excludes exposed high voltage?	Source-based mechanism map.

Three visual chains to redraw from memory



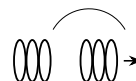
Charge → field → force

Lesson 1 establishes interaction; Lesson 2 maps how a test object responds by location.



Current → field → induction

Lessons 3–5 connect a measured current to a field effect and then to change-driven induction.



Ratio → resonance → coupling

Lessons 6–9 test how turns, frequency, separation, and orientation shape transfer.

CHECKPOINT A — AFTER LESSON 3

Without looking back, draw a source, a load, meter placements, and conventional current. Predict the sign and approximate scale of one reading. Continue only after the diagram and physical setup agree.

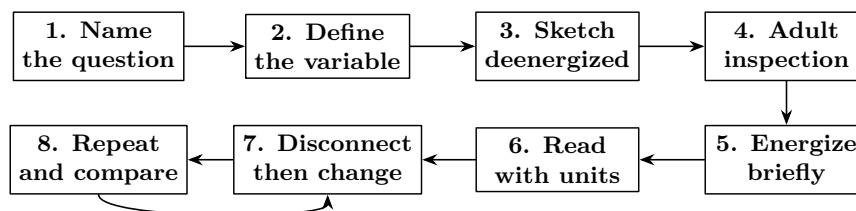
CHECKPOINT B — AFTER LESSON 6

Given a doubled secondary turn count, predict whether secondary voltage should rise or fall, then state two reasons a real reading may miss the ideal ratio.

CHECKPOINT C — AFTER LESSON 9

Explain the complete chain from source energy to a received meter response, naming every conversion and one measurement that tests each link.

A repeatable evidence workflow



Before the first trial

Question	What relationship are we testing?
Prediction	If _____ changes, then _____, because _____.
Independent variable	The one quantity deliberately changed, with planned values.
Response variable	What will be observed or measured, including unit and range.
Controls	Geometry, parts, timing, meter range, preparation method, and conditions held fixed.
Stop condition	Heat, odor, damage, pain, unstable connection, unexpected behavior, or adult decision.

Trial table — never separate a number from its conditions

Trial	Changed quantity + unit	Response + unit	Conditions	Observation / anomaly
1				
2				
3				
4				
5				

Reading the result

Support. The repeated observations change in the predicted direction, and the effect is larger than the reading resolution or ordinary scatter.

Refine. The direction agrees but the size does not. Check idealized assumptions, meter loading, resistance, geometry, timing, and uncertainty.

Reject for now. The pattern is absent or reverses. Preserve the result; check the setup and prediction without quietly deleting inconvenient trials.

Inconclusive. The effect is comparable to noise, a control changed, or the apparatus could not be inspected. Improve the test before making a claim.

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DIAGNOSTIC PAUSE

If two trials disagree, do not add energy or exceed the boundary. Disconnect. Check the drawing against the setup; confirm meter mode and range; inspect each connection; repeat a known baseline; then alter only the planned variable.

Predict first

Learner questions. What result would change your mind? Which control is most likely to drift? What is the smallest change your instrument can show? Could another mechanism produce the same observation? What should a second person be able to reproduce from your record alone?

The bound notebook

Number every page. Never remove a failed trial. Begin each lesson on a new spread and give every sketch, table, and graph a stable identifier such as L04-F2. Cross-reference revisions rather than erasing the earlier model. Paste or transcribe meter specifications and part values beside the first use.

LEFT: plan and apparatus

- question + prediction
- variable table
- dated circuit or geometry sketch
- adult inspection initials

RIGHT: evidence and judgment

- raw readings with units
- graph or comparison
- anomalies retained
- claim, limitation, next test

_____ references tie claim to evidence →

Required record for each lesson	
Header	Date; lesson; partners; apparatus ID; source voltage; meter and range.
Question	One relationship or mechanism phrased so evidence can answer it.
Prediction	Direction, approximate size when possible, and physical reason.
Plan	Changed quantity, response quantity, controls, sequence, stop condition.
Apparatus record	Deenergized sketch with component values, polarity, meter placement, distances, and orientations.
Raw evidence	Every trial in acquisition order, units in headings, resolution, and anomalies.
Analysis	Calculation or graph; comparison to prediction; uncertainty or scatter.
Claim	One bounded sentence followed by specific evidence: “because trials ...”.
Revision	What changed in the model, what remains unexplained, and the next discriminating test.
Sign-off	Learner check; adult inspection; explicit confirmation that the boundary was maintained.

Course-level index

L.	Record IDs	Main measured pat-tern	Model revision	Open question / next link
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

COURSE COMPLETION CHECK

Choose one claim from Lessons 1–3, one from 4–6, and one from 7–9. For each, locate the raw observation, reconstruct the apparatus from the drawing, state the main uncertainty, and name a result that would have weakened the claim. Lesson 10 is complete only when the learner can explain the historical mechanism and the reason it remains outside learner construction.

Demonstrations and provider independence

Use a demonstration only after its named prerequisite and record it as observation, not as a substitute for the learner investigation. Finished prose and artwork in this edition are independently authored. The shared neutral research library supplies authoritative physics and safety claims; full citations and scope notes are recorded there.

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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